

Resource and Trajectory Optimization for UAV-Relay-Assisted Secure Maritime MEC

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Abstract—With the evolutionary development of maritime networks, the explosive growth of maritime data has put forward elevated demands for the computing capabilities of maritime devices (MDs). Unmanned aerial vehicle (UAV) is able to alleviate the computing pressure of MDs by forwarding the computing tasks to the edge server on the coast. However, UAV relaying introduces a significant security challenge due to the vulnerability of line-of-sight (LoS) communication channels, which can be exploited for eavesdropping on computing tasks. In this paper, an efficient secure communication scheme is proposed for UAV-relay-assisted maritime mobile edge computing (MEC) with a flying eavesdropper. The secure computing capacity of MDs is maximized by jointly optimizing the transmit power, time slot allocation factor, computation optimization and UAV trajectory. Due to multi-variable coupling, the formulated optimization problem (OP) is non-convex. We first transform OP by introducing auxiliary variables. Then, the transformed OP is decomposed and solved in an iterative manner by applying block coordinate descent (BCD) and successive convex approximation (SCA). Numerical results show that the secure computing capability of the UAV-relay-assisted maritime MEC system of proposed secure communication scheme can be effectively improved compared with benchmarks.

Index Terms—Secure transmission, MEC, UAV-relay, maritime communications.

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I. INTRODUCTION

WITH the evolutionary development of technological advancements, the maritime services are becoming more and more important, e.g., maritime fishery, transportation, search and rescue, and military activities. To better realize miscellaneous maritime services, it is urgent to establish reliable links between maritime devices (MDs) and the coast [1]. The explosive growth of massive maritime data will put forward higher requirements for the computing capabilities of MDs [2]. Through offloading the computation tasks to the edge servers with more powerful capability deployed on the coast, mobile edge computing (MEC) has the capability to effectively alleviate the computation pressure of MDs [3]. However, most of MDs are deployed far from the coast. Limited by the maritime infrastructure and poor maritime channels, the direct communication between MDs and the coastal edge server (CES) is always difficult to support the transmission of massive computing tasks.

Taking advantage of the inherent characteristics of high mobility and flexibility [4], [5], unmanned aerial vehicles (UAVs) can be utilized to enhance the transmission quality between MDs and CES by serving as a mobile relay [6]. Through the optimization of both relay trajectory and power, Zeng et al. utilized the controllable channel changes caused by relay movement under the constraints of UAV movement and information causality, leading to an improvement in end-to-end throughput [7]. Chen et al. analyzed the interrupt performance and bit error rate of single and multi-hop links in multiple UAVs relaying systems through designing the optimal placement of UAVs [8]. In [9], Zhao et al. presented a framework that utilizes UAVs to establish information exchange via multi-hop UAV relaying. Zhong et al. in [10] handled uncertain channel models and communication node locations for the UAV-assisted-relaying networks. Liang et al. investigated a UAV-assisted bidirectional relaying scheme that involved multiple ground user pairs [11]. Zhang et al. utilized UAV-assisted decode-and-forward relaying by adjusting placement of the UAV to enhance the caching performance [12]. Zhan et al. minimized the completion time and energy consumption of the UAV-enabled MEC system by jointly designing the computation offloading, resource allocation and UAV trajectory [13].

With the assistance of UAV relaying, the flexibility of MEC can be enhanced [14]. Specifically, UAV can facilitate the offloading of computing tasks to the CES that possesses

superior computational capabilities [15]. Hu et al. studied the effectiveness of UAV-assisted MEC with various constraints, such as task bandwidth allocation, information causality, and UAV trajectory [16]. Liu et al. maximized the energy-efficiency of the maritime system by jointly optimizing the UAV's trajectory and the individual transmit power levels of the source and the UAV relay nodes [17]. Na et al. investigated a high-efficiency scheme for UAV relaying communication maritime system with the UAV trajectory and resource allocation optimization [18]. Zhang and Ansari jointly optimized the UAV deployment, terminal association, time allocation of access and backhaul links, computing resource distribution to minimize the average delay of terminals, with UAVs serving as computing and relaying nodes [19]. Liu et al. investigated a multi-input single-output UAV-assisted MEC scheme that addressed the challenges of poor channel quality due to multi-path and blocking in traditional MEC networks [20]. He et al. in [21] proposed a scheme for multi-hop task offloading based on dynamic computing that aimed to achieve more powerful remote edge computing using multiple UAVs. Zhao et al. in [22] presented an optimization framework that involves UAV-assisted vehicle computing and offloading, in which the offloading decision problem was transformed into a multi-player computational offloading sequence game problem.

Nevertheless, due to the line-of-sight (LoS) characteristics, the offloaded data of UAV-assisted MEC networks can be easily eavesdropped, causing serious security threat [23]. Physical layer security enables the achievement of secure offloading, protecting the data against malicious eavesdropping [24], [25]. Wang et al. investigated transmission optimization in a four-node system with the goal of maximizing the secure rate, and proposed an iterative algorithm based on the difference-of-concave method [26]. Zhou et al. conducted a joint secure transmission optimization of UAV position, computing power, user association, transmit power and offloading ratio with multiple eavesdropping UAVs [27]. Li et al. investigated a energy-saving scheme for UAV-MEC secure transmissions, where the optimization of transmit power, task allocation, and UAV placement was performed with considering secure offloading rate constraints [28]. Xu et al. employed a two-UAV framework to facilitate secure transmissions with the existence of multiple ground eavesdroppers, where one UAV was utilized for offloading tasks, while the other was dedicated to mitigating the risk of malicious eavesdropping [29]. Meanwhile, a secure computation scheme was investigated by Lu et al. to overcome the secure threat in UAV-assisted MEC, where one UAV offering computing services to the users and another UAV eavesdropping on the transmission of their data [30], [31]. Comparing with the ground eavesdroppers, which are deployed at the fixed positions in the existing works, UAV eavesdroppers will have much better channel condition. Thus, the information can be easily eavesdropped by flying UAVs. The major challenges for considering UAV eavesdroppers are to consider the anti-collision constraint between UAVs, which is not considered for ground eavesdroppers.

However, the secure transmission in UAV-relay-assisted maritime MEC has not been well studied in existing works.

Thus, an effective secure communication scheme is proposed in this paper for the UAV-relay-assisted maritime MEC network. Specifically, a UAV relay (UAV_r) helps MDs offload tasks to a CES, while a UAV eavesdropper (UAV_e) eavesdrops the offloaded data. To prevent eavesdropping, a coastal jammer (CJ) generates the jamming signals to disrupt UAV_e eavesdropping. UAV_r trajectory, time slot allocation, transmit power and computing assignment are optimized to maximize the minimum secure computing capacity of MDs. The contributions are summarized as follows.

- Due to the LoS transmission, the secure transmission remains as a challenging issue in UAV-relay assisted maritime MEC networks, which has not been well studied in the existing works. Thus, we propose a secure communication scheme for UAV-relay-assisted maritime MEC networks.
- In the proposed scheme, UAV_r helps MDs to offload computing tasks to CES. CJ emits the jamming signals to disrupt UAV_e eavesdropping. Considering the constraints of the flight speed and anti-collision of UAV, the transmit power of MDs and UAV_r , the local computing capacity of MDs, an optimization problem (OP) is formulated to maximize the minimum secure computing capacity by optimizing the transmit power, time slot allocation factor, computation allocation and UAV trajectory.
- To tackle this multivariate coupling OP, a block coordinate descent (BCD) based joint optimization algorithm is proposed. Specifically, we first introduce auxiliary variables to facilitate the transformation of OP into an equivalent form. Subsequently, OP is decomposed into a set of subproblems. Then, based on SCA, the approximate solution is efficiently obtained in an iterative way. Furthermore, the feasibility and complexity of the proposed scheme are discussed to prove the effectiveness.

The rest of the paper is organized as follows. The UAV-relay-assisted secure maritime MEC system is introduced in Section II. Section III formulates the problem of maximizing secure computing capacity. A BCD-based joint optimization algorithm is studied in Section IV to solve OP. Numerical results are shown and analyzed in Section V. Section VI concludes the paper.

II. SYSTEM MODEL

A UAV-relay-assisted secure maritime MEC system is considered as shown in Fig. 1. There are K MDs, two UAVs (UAV_r and UAV_e), one CES and one CJ. UAV_r forwards the computing tasks of MDs to CES through amplify-and-forward (AF) relaying. UAV_e eavesdrops the transmission from MDs to UAV_r . CJ sends the jamming signal to disrupt the eavesdropping. We assume that CES has prior knowledge of the jamming signal sent by CJ because CES and CJ belong to the legitimate network, the jamming signal sent by CJ are friendly to CES. Thus, CES will not be affected by the artificial jamming signals. However, UAV_e is unaware of CJ's presence because UAV_e is a mobile eavesdropper and it does not belong to the legitimate network. We assume that UAV_r and CES already know the channel state information among CES, CJ,

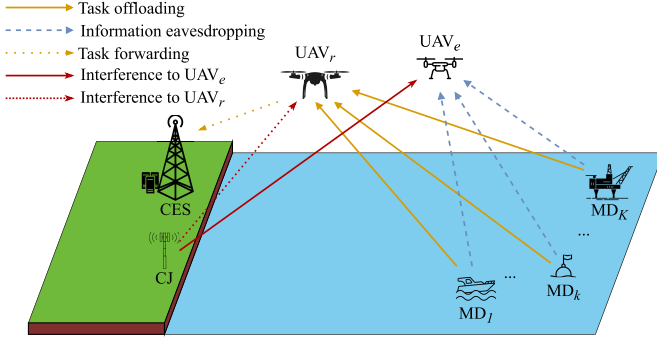


Fig. 1. UAV-relay-assisted secure maritime MEC system.

TABLE I
NOTATIONS

Notation	Description
$\mathbf{v}_k$	Location of MD _k
$\mathbf{v}_s$	Location of CES
$\mathbf{v}_j$	Location of CJ
$\mathbf{u}_i[n]$	Location of UAV _i
H_i	Altitude of UAV _i
φ_t	Length of the time slot
T	Flight period of UAVs
N	Number of total time slots
$\mathbf{u}_i^I$	Initial position of UAV _i
$\mathbf{u}_i^F$	Final position of UAV _i
$V_i^{\max}$	Maximum speed of UAV _i
$d_{\min}$	Minimum distance between UAVs
$\tilde{G}[n]$	Power ratio of the signal
β_0	Channel power gain at a unit distance
$P_k^{\max}$	Peak power of MD _k
$P_r^{\max}$	Peak power of UAV _r
$p_k[n]$	Transmit power of MD _k
$p_r[n]$	Transmit power of UAV _r
$\xi_k[n]$	Time slot allocation factor
$x_k[n]$	Transmit signal of MD _k
$x_j[n]$	Transmit signal of CJ
n_r	Noise power of UAV _r
n_e	Noise power of UAV _e
n_s	Noise power of CES
c_k	CPU cycles for MD calculating per bit
c_s	CPU cycles for CES calculating per bit
$l_{k,\text{loc}}[n]$	Local computing allocation of MD _k
$F_k^{\max}$	Peak CPU frequency of MD
$F_s^{\max}$	Peak CPU frequency of CES
W	Communication bandwidth
Q_k	Secure computing requirement of MD _k
k_k	CPU capacity coefficient of MDs
P_k^{ave}	Power budget of MD _k

UAV_r and MDs in advance by means of synthetic aperture radar, etc. The symbols used are shown in Table I.

The horizontal coordinates of MD_k, CES and CJ are described as $\mathbf{v}_{u_0} = (x_{u_0}, y_{u_0})^T$, $u_0 \in \{k, s, j\}$, where $k \in \mathcal{K} = \{1, 2, \dots, K\}$. The flight period of UAVs is given by T , $T \geq 0$. The position of UAV_i at time t is denoted as $\mathbf{u}_i(t)$, $i \in \{r, e\}$, $t \in [0, T]$. Assume that the altitude of UAV_i is constant, denoted by H_i . To simplify the discussion, T is uniformly partitioned into N discrete time slots. Thus, the length of a single slot is $\varphi_t = T/N$. In slot n , the position of UAV_i is denoted as $\mathbf{u}_i[n] = (x_i[n], y_i[n])^T$.

Assume that UAV_i flies from the setting initial position $\mathbf{u}_i^I$ to the setting final position $\mathbf{u}_i^F$ within the period T . The flight speed of UAV_r cannot exceed $V_r^{\max}$. Then, the

maximum allowable displacement of UAV_r in a single time slot is $l_r^{\max} = V_r^{\max} \varphi_t$. Then, we have

$$\mathbf{u}_r[1] = \mathbf{u}_r^I, \quad (1a)$$

$$\mathbf{u}_r[N] = \mathbf{u}_r^F, \quad (1b)$$

$$\|\mathbf{u}_r[n+1] - \mathbf{u}_r[n]\| \leq l_r^{\max}. \quad (1c)$$

To avoid collision of UAV_r and UAV_e, the minimum collision avoidance distance between them is $d_{\min}$. We have

$$\|\mathbf{u}_r[n] - \mathbf{u}_e[n]\|^2 \geq d_{\min}^2. \quad (2)$$

In this work, we use distance-dependent path loss model [29]. In slot n , the distances between MD_k and UAV_r, MD_k and UAV_e, CJ and UAV_r, CJ and UAV_e, UAV_r and CES can be respectively expressed as

$$d_{u_1, u_2}[n] = \sqrt{H_{u_2}^2 + \|\mathbf{u}_{u_2} - \mathbf{v}_{u_1}\|^2}, \quad (3)$$

where $(u_1, u_2) \in \{(k, r), (k, e), (j, r), (j, e), (s, r)\}$.

Since CES and CJ are deployed on the coast, we consider them to be approximated as maritime nodes. Thus, the channels between MD_k and UAV_r, MD_k and UAV_e, CJ and UAV_r, CJ and UAV_e, UAV_r and CES can be considered as air-to-sea channels. They follow Rician fading, which can be taken as composite channels of large-scale fading and small-scale fading [32], [33]. Thus, the corresponding channel coefficients can be expressed as

$$\begin{aligned} h_{u_1, u_2}[n] &= g_{u_1, u_2}[n] \tilde{h}_{u_1, u_2}[n] \\ &= \sqrt{\frac{\beta_0}{d_{u_1, u_2}^2[n]}} \left(\sqrt{\frac{\tilde{G}[n]}{\tilde{G}[n]+1}} + \sqrt{\frac{1}{\tilde{G}[n]+1}} \tilde{g}_{u_1, u_2}[n] \right)^2, \end{aligned} \quad (4)$$

where $\{g_{u_1, u_2}[n]\}$ and $\{\tilde{h}_{u_1, u_2}[n]\}$ are the large-scale fading coefficients and small-scale fading coefficients, respectively. β_0 is channel power gain at a unit distance on the LoS component of the signal, and $\tilde{G}[n]$ is the power ratio of the LoS to the non-line-of-sight (NLoS) signal. $\tilde{g}_{u_1, u_2}[n] \in \mathcal{CN}(0, 1)$ is the ratio of the NLoS component matrix between the LoS component matrix of the signal.

Define $P_k^{\max}$ and $P_r^{\max}$ as the peak power of MD_k and UAV_r, respectively. The transmit power of MD_k and UAV_r need to satisfy

$$0 \leq p_k[n] \leq P_k^{\max}, \quad \forall k, n, \quad (5a)$$

$$0 \leq p_r[n] \leq P_r^{\max}, \quad \forall k, n. \quad (5b)$$

III. PROBLEM FORMULATION

In the previous section, the flight period T is uniformly partitioned into N discrete time slots based on the flight trajectory of UAVs. In each slot, every MD can access the UAV_r for task forwarding and offloading. To avoid the communication interference between MDs, task offloading is performed using time-division-multiple-access, where one slot is further partitioned into K subslots. The length of time

allocated to MD_k is $\xi_k[n]\varphi_t$, where $\xi_k[n]$ is time slot allocation factor. $\xi_k[n]$ is limited by

$$\sum_{k=1}^K \xi_k[n] \leq 1, \quad \forall n, \quad (6a)$$

$$0 \leq \xi_k[n] \leq 1, \quad \forall k, n. \quad (6b)$$

A. Communication Model

Each subslot $\xi_k[n]\varphi_t$ is further divided into two phases with equal length of time.

In the first phase, MD_k offloads its computing task to UAV_r , which will be eavesdropped by UAV_e . CJ transmits signals to interfere the eavesdropping of UAV_e . Thus, the signal received in the first phase of UAV_r is

$$y_{k,r}^{(1)}[n] = \sqrt{p_k[n]}h_{k,r}[n]x_k[n] + \sqrt{P_j}h_{j,r}[n]x_j[n] + n_r, \quad \forall k, n, \quad (7)$$

where P_j is the transmit power of CJ, $x_k[n]$ and $x_j[n]$ is the information sent by MD_k and CJ, respectively, and n_r is the received noise of UAV_r .

Assume that $x_k[n]$ and $x_k[j]$ are both normalized,

$$\mathbb{E}\{\|x_k[n]\|^2\} = \mathbb{E}\{\|x_j[n]\|^2\} = 1, \quad \forall n. \quad (8)$$

When UAV_e eavesdrops the transmission from MD_k to UAV_r in the first phase, it also receives the interfering signal transmitted by CJ. Thus, the signal received by UAV_e is

$$y_{k,e}^{(1)}[n] = \sqrt{p_k[n]}h_{k,e}[n]x_k[n] + \sqrt{P_j}h_{j,e}[n]x_j[n] + n_e, \quad \forall k, n, \quad (9)$$

where n_e is the received noise of UAV_e .

As UAV_e cannot distinguish the interfering signal transmitted by CJ, the SINR of UAV_e eavesdropping on the signal from MD_k to UAV_r can be expressed as

$$\gamma_{k,e}[n] = \frac{|h_{k,e}[n]|^2 p_k[n]}{|h_{j,e}[n]|^2 P_j + n_e^2}, \quad \forall k, n. \quad (10)$$

In the second phase, UAV_r forwards the information to CES via AF. The amplification factor is

$$G_{k,r}[n] = \sqrt{\frac{p_r[n]}{|h_{k,r}[n]|^2 p_k[n] + |h_{j,r}[n]|^2 P_j + n_r^2}}, \quad \forall k, n, \quad (11)$$

where $p_r[n]$ is UAV_r 's transmit power.

Then, the signal received by CES is

$$y_{r,s}^{(2)}[n] = G_{k,r}[n]h_{r,s}[n]y_{k,r}^{(1)}[n] + n_s, \quad (12)$$

where n_s is the received noise of CES.

Since CES knows the interfering signal of CJ in advance, the SINR of CES from MD_k is written as (13), shown at the bottom of the next page.

Therefore, the task offloading rate from MD_k to CES is

$$\Phi_{k,s}[n] = \log_2(1 + \gamma_{k,s}[n]), \quad \forall k, n. \quad (14)$$

The eavesdropping rate from MD_k to UAV_e is

$$\Phi_{k,e}[n] = \log_2(1 + \gamma_{k,e}[n]), \quad \forall k, n. \quad (15)$$

Then, the secure offloading rate from MD_k to CES is

$$\Phi_{k,\text{sec}}[n] \triangleq (\Phi_{k,s}[n] - \Phi_{k,e}[n])^+. \quad (16)$$

B. Computing Model

A partial offload strategy is adopted at MDs, in which, a part of tasks are executed on MDs, and the other tasks are forwarded to CES through UAV_r for execution.

Assume that the number of CPU cycles for MD_k when calculating a single bit data are c_k . The amount of data computed locally by MD_k in slot n are $l_{k,\text{loc}}[n]$. Assume that MD_k 's maximum CPU frequency is $F_k^{\max}$. Thus, we have

$$c_k l_{k,\text{loc}}[n] \leq \varphi_t F_k^{\max}, \quad \forall k, n. \quad (17)$$

Define the CPU cycles of CES to calculate a single bit data as c_s , and CES's maximum CPU frequency as $F_s^{\max}$. Thus, the number of bits computed on CES cannot exceed its computing capacity, satisfying

$$\frac{1}{2} c_s W \xi_k[n] \varphi_t \Phi_{k,\text{sec}}[n] \leq \frac{1}{2} \varphi_t \xi_k[n] F_s^{\max}, \quad (18)$$

where W is bandwidth.

Define the minimum secure computing requirement of MD_k as Q_k . To guarantee the secure computing requirement of each MD, we have

$$l_{k,\text{loc}}[n] + \frac{1}{2} W \xi_k[n] \varphi_t \Phi_{k,\text{sec}}[n] \geq Q_k, \quad \forall k, n. \quad (19)$$

If tasks are computed locally at MD_k , the energy consumption can be written as

$$E_{k,\text{loc}}[n] = \frac{k_k (c_k l_{k,\text{loc}}[n])^3}{\varphi_t^2}, \quad (20)$$

where k_k denotes the effective capacitance coefficient of MD_k .

On the other hand, if tasks are computed remotely at CES, the energy of MD_k is consumed at transmitting

$$E_{k,\text{trans}}[n] = \frac{1}{2} \xi_k[n] \varphi_t p_k[n]. \quad (21)$$

The energy consumption of MD_k in the whole period T should satisfy

$$\frac{1}{T} \sum_{n=1}^N (E_{k,\text{loc}}[n] + E_{k,\text{trans}}[n]) \leq P_k^{\text{ave}}, \quad \forall k, \quad (22)$$

where P_k^{ave} is the average power limit of MD_k .

C. Problem Formulation

The secure computing capacity of MD_k is defined as the average number of bits that MD_k can achieve, which is composed of local and offloading computation,

$$\bar{\Phi}_{k,\text{sec}} \triangleq \frac{1}{T} \left(\frac{1}{2} W \varphi_t \sum_{n=1}^N \xi_k[n] \Phi_{k,\text{sec}}[n] + \sum_{n=1}^N l_{k,\text{loc}}[n] \right), \quad \forall k. \quad (23)$$

Our optimization target is to maximize the minimum secure computing capacity of the system by optimizing the time slot allocation factor $\xi_k[n]$, MD_k transmit power $p_k[n]$, UAV_r transmit power $p_r[n]$, the amount of data computed at MD_k

locally $l_{k,\text{loc}}[n]$ and the UAV_r trajectory $\mathbf{u}_r[n]$, which can be formulated as

$$(P1) : \begin{aligned} & \max_{\{\xi_k[n], p_k[n], p_r[n], l_{k,\text{loc}}[n], \mathbf{u}_r[n]\}} \min_{\forall k} \bar{\Phi}_{k,\text{sec}} \\ & \text{s.t. (1),(2),(5),(6),(17)-(22),} \end{aligned} \quad (24)$$

Due to the coupling of multi-variables, the non-convexity of constraints (2), (18), (19) and (22), the original problem (P1) is non-convex. Specifically, the constraint (2) is non-convex. Constraints (18) and (19) are related to $\Phi_{k,\text{sec}}[n]$, which is composed of multiple optimization variables, e.g., $p_k[n]$, $p_r[n]$, $\mathbf{u}_r[n]$, making (18) and (19) multi-variable coupled and non-convex. Moreover, the constraint (22) is also related to multiple optimization variables, e.g., $\xi_k[n]$, $p_k[n]$, $l_{k,\text{loc}}[n]$, making (22) non-convex.

IV. PROBLEM SOLUTION

To simplify the resolution of the problem (1), we transform it into an equality form with three auxiliary variables θ , $\theta_{1,k}[n]$ and $\theta_{2,k}[n]$

$$(P2) : \begin{aligned} & \max_{\mathcal{Z}} \theta \\ & \text{s.t. (1), (2), (5), (6), (17), (22),} \end{aligned} \quad (25a)$$

$$\theta \leq \frac{1}{T} \left(\frac{1}{2} W \varphi_t \sum_{n=1}^N \xi_k[n] (\theta_{1,k}[n] - \theta_{2,k}[n]) + \sum_{n=1}^N l_{k,\text{loc}}[n] \right), \quad \forall k, \quad (25b)$$

$$\theta_{1,k}[n] \leq \Phi_{k,s}[n], \quad \forall k, n, \quad (25c)$$

$$\theta_{2,k}[n] \geq \Phi_{k,e}[n], \quad \forall k, n, \quad (25d)$$

$$c_s W (\theta_{1,k}[n] - \theta_{2,k}[n]) \leq F_s^{\max}, \quad \forall k, n, \quad (25e)$$

$$l_{k,\text{loc}}[n] + \frac{1}{2} B \xi_k[n] \varphi_t (\theta_{1,k}[n] - \theta_{2,k}[n]) \geq Q_k, \quad \forall k, n, \quad (25f)$$

where $\mathcal{Z} = \{\xi_k[n], p_k[n], p_r[n], l_{k,\text{loc}}[n], \mathbf{u}_r[n], \theta_{1,k}[n], \theta_{2,k}[n]\}$.

For (18), (19) and (23), by setting $p_k[n] = 0$, $p_r[n] = 0$ and $l_{k,\text{loc}}[n] = 0$, at least the value of zero can be obtained. Thus, we can omit the operator $[\cdot]^+$. We introduce the auxiliary variable s as the lower bound of $\bar{\Phi}_{k,\text{sec}}$ as shown in (25b). Meanwhile, $\theta_{1,k}[n]$ is introduced to represent the lower bound of $\Phi_{k,s}[n]$ and $\theta_{2,k}[n]$ is introduced to represent the upper bound of $\Phi_{k,e}[n]$, as shown in (25c) and (25d), respectively. The value of θ can be always enlarged, unless the equality in (25b) is hold at the optimal solution. Similarly, in order to achieve the optimal solution, it is necessary for at least one equality in equations (25c) and (25e) to hold, thereby ensuring the same value as the problem (P1). Therefore, in the case of $\theta_{2,k}[n]$, the equality $\theta_{2,k}[n] = \max \Phi_{k,e}[n]$ must hold at the optimal solution. Otherwise, if $\theta_{2,k}[n]$ is not equal to $\max \Phi_{k,e}[n]$, it can always be decreased, leading to a larger

value of the objective function. Therefore, the transformed problem (P2) is equivalent to (P1).

To overcome the problem (P2), a BCD-based joint optimization algorithm is proposed. We adopt a two-step approach with block structures of the variables. In Step 1, the variables $\mathcal{Z} \setminus \mathbf{u}_r[n]$ are optimized by considering $\mathbf{u}_r[n]$ at a fixed value. Subsequently, in Step 2, the variables $\mathbf{u}_r[n]$ is optimized by considering $\mathcal{Z} \setminus \mathbf{u}_r[n]$ fixed.

A. Step 1: Optimizing $\mathcal{Z} \setminus \mathbf{u}_r[n]$

For fixed $\mathbf{u}_r[n]$, the problem (P2) can be re-expressed as

$$(P3) : \begin{aligned} & \max_{\mathcal{Z}} \theta \\ & \text{s.t. (5),(6),(17),(22),(25b)-(25f).} \end{aligned} \quad (26)$$

The problem (P3) has non-convex constraints, which are shown as (22), (25b)-(25d), and (25f). As a result, the problem (P3) is also non-convex. To overcome this non-convexity, we can utilize the BCD to solve [34]. Specifically, we can obtain the time slot allocation factor of MD_k $\xi_k[n]$, MD_k transmit power $p_k[n]$, UAV_r transmit power $p_r[n]$, and MD_k local computation allocation $l_{k,\text{loc}}[n]$ by fixing the other values in an iterative manner.

1) *Time Allocation*: For the fixed MD_k transmit power $p_k[n]$, UAV_r transmit power $p_r[n]$ and MD_k local computation allocation $l_{k,\text{loc}}[n]$, the OP (P3) is

$$(P3.1) : \begin{aligned} & \max_{\{\xi_k[n], \theta_{1,k}[n], \theta_{2,k}[n]\}} \theta \\ & \text{s.t. (6),(22),(25b)-(25f).} \end{aligned} \quad (27)$$

(P3.1) is a typical convex OP, which is solved using traditional convex optimization methods such as CVX [35].

2) *MD_k Transmit Power*: For the fixed time slot allocation factor $\xi_k[n]$, UAV_r transmit power $p_r[n]$ and MD_k local computation allocation $l_{k,\text{loc}}[n]$, the OP (P3) is formulated as

$$(P3.2) : \begin{aligned} & \max_{\{p_k[n], \theta_{1,k}[n], \theta_{2,k}[n]\}} \theta \\ & \text{s.t. (5a),(22),(25b)-(25f).} \end{aligned} \quad (28)$$

Because of (25c) and (25d)'s non-convexity, (P3.2) is difficult to solve. The OP (P3.2) can be solved by SCA [36], where problem (P3.2) can be approximated as the convex problem in each iteration. The transmit power of MD_k is obtained via iterations.

Assume that $p_k^{(i)}[n]$ is the value of the i th iteration of the transmit power $p_k[n]$ of MD_k. By applying SCA, we take the first-order Taylor expansion at the given $p_k^{(i)}[n]$, and convert (25c) to

$$\theta_{1,k}[n] \leq \log_2 \left(1 + \frac{a_1 p_k^{(i)}[n]}{b_1 p_k^{(i)}[n] + c_1} \right) + \frac{a_1 c_1}{\ln 2} \frac{p_k[n] - p_k^{(i)}[n]}{\left((a_1 + b_1) p_k^{(i)}[n] + c_1 \right) \left(b_1 p_k^{(i)}[n] + c_1 \right)}, \quad (29)$$

$$\gamma_{k,s}[n] = \frac{p_k[n] |h_{k,r}[n]|^2 p_r[n] |h_{r,s}[n]|^2}{p_r[n] |h_{r,s}[n]|^2 n_r^2 + p_k[n] |h_{k,r}[n]|^2 n_s^2 + P_j |h_{j,r}[n]|^2 n_s^2 + n_r^2 n_s^2}, \quad \forall k, n \quad (13)$$

where

$$\begin{cases} a_1 = |h_{k,r}[n]|^2 p_r[n] |h_{r,s}[n]|^2, \\ b_1 = |h_{k,r}[n]|^2 n_s^2, \\ c_1 = p_r[n] |h_{r,s}[n]|^2 n_r^2 + P_j |h_{j,r}[n]|^2 n_s^2 + n_r^2 n_s^2. \end{cases} \quad (30)$$

(25d) can be similarly converted to

$$\theta_{2,k}[n] \geq \log_2 \left(1 + \frac{a_2 p_k^{(i)}[n]}{b_2} \right) + \frac{a_2}{\ln 2} \frac{p_k[n] - p_k^{(i)}[n]}{a_2 p_k^{(i)}[n] + b_2}, \quad \forall k, n, \quad (31)$$

where

$$\begin{cases} a_2 = |h_{k,e}[n]|^2, \\ b_2 = |h_{j,e}[n]|^2 P_j + n_e^2. \end{cases} \quad (32)$$

Thus, the problem (P3.2) can be reformulated as

$$\begin{aligned} \text{(P3.2.1)} : \quad & \max_{\{p_k[n], \theta_{1,k}[n], \theta_{2,k}[n]\}} \theta \\ & \text{s.t. (5a), (22), (25b), (25e), (25f), (29), (31).} \end{aligned} \quad (33)$$

Note that (P3.2.1) is convex, which can be solved through CVX.

3) *UAV_r Transmit Power*: For the fixed time slot allocation factor $\xi_k[n]$, MD_k transmit power $p_k[n]$ and MD_k local computation allocation $l_{k,\text{loc}}[n]$, the OP (P3) can be formulated as

$$\begin{aligned} \text{(P3.3)} : \quad & \max_{\{p_r[n], \theta_{1,k}[n], \theta_{2,k}[n]\}} \theta \\ & \text{s.t. (5b), (25b)-(25f).} \end{aligned} \quad (34)$$

The problem (P3.3) is difficult to solve as (25c) is non-convex. Similarly, we can use SCA to solve it. Assume that $p_r^{(i)}[n]$ is the value of the i_{th} iteration of the transmit power $p_r[n]$ of UAV_r. With the given point $p_r^{(i)}[n]$, through taking the first-order of Taylor expansion, we can convert (25c) as

$$\begin{aligned} & \theta_{1,k}[n] \\ & \leq \log_2 \left(1 + \frac{a_3 p_r^{(i)}[n]}{b_3 p_r^{(i)}[n] + c_3} \right) \\ & + \frac{a_3 c_3}{\ln 2} \frac{p_r[n] - p_r^{(i)}[n]}{\left((a_3 + b_3) p_r^{(i)}[n] + c_3 \right) \left(b_3 p_r^{(i)}[n] + c_3 \right)}, \quad \forall k, n, \end{aligned} \quad (35)$$

where

$$\begin{cases} a_3 = p_k[n] |h_{k,r}[n]|^2 |h_{r,s}[n]|^2, \\ b_3 = |h_{r,s}[n]|^2 n_r^2, \\ c_3 = p_k[n] |h_{k,r}[n]|^2 n_s^2 + P_j |h_{j,r}[n]|^2 n_s^2 + n_r^2 n_s^2. \end{cases} \quad (36)$$

Then, the problem (P3.3) can be reformulated as

$$\begin{aligned} \text{(P3.3.1)} : \quad & \max_{\{p_r[n], \theta_{1,k}[n], \theta_{2,k}[n]\}} \theta \\ & \text{s.t. (5b), (25b), (25d)-(25f), (35).} \end{aligned} \quad (37)$$

The constraints of the problem (P3.3.1) are all linear. Thus, the problem (P3.3.1) is typically convex, which can be solved by CVX.

4) *MD_k Local Computation Allocation*: For the fixed time slot allocation factor $\xi_k[n]$, MD_k transmit power $p_k[n]$ and UAV_r transmit power $p_r[n]$, the OP (P3) can be formulated as

$$\begin{aligned} \text{(P3.4)} : \quad & \max_{\{l_{k,\text{loc}}[n], \theta_{1,k}[n], \theta_{2,k}[n]\}} \theta \\ & \text{s.t. (17), (22), (25b)-(25f).} \end{aligned} \quad (38)$$

As the constraint (22) is convex, and the rest constraints are linear, (P3.4) can be solved by CVX.

B. Step 2: Optimizing $\mathbf{u}_r[n]$

By fixing the time slot allocation factor $\xi_k[n]$, MD_k transmit power $p_k[n]$, UAV_r transmit power $p_r[n]$, and MD_k local computation allocation $l_{k,\text{loc}}[n]$, we can reformulate the problem (P2) as

$$\begin{aligned} \text{(P4)} : \quad & \max_{\{\mathbf{u}_r[n], \theta_{1,k}[n], \theta_{2,k}[n]\}} \theta \\ & \text{s.t. (1), (2), (25b)-(25f),} \end{aligned} \quad (39)$$

where the constraints (2) and (25c) are non-convex.

Similarly, we can use SCA to solve it. Assume that $\mathbf{u}_r^{(i)}[n]$ is the value of the i_{th} iteration of the trajectory of UAV_r, through taking the first-order Taylor expansion, (2) and (25c) are converted to

$$\begin{aligned} & \|\mathbf{u}_r^{(i)}[n] - \mathbf{u}_e[n]\|^2 \\ & + 2(\mathbf{u}_r^{(i)}[n] - \mathbf{u}_e[n]) \cdot (\mathbf{u}_r[n] - \mathbf{u}_r^{(i)}[n]) \geq d_{\min}^2, \quad \forall k, n. \end{aligned} \quad (40)$$

$$\theta_{1,k}[n] \leq f_{1,T}(\mathbf{u}_r[n]) - f_{2,T}(\mathbf{u}_r[n]), \quad \forall k, n, \quad (41)$$

where $f_{\alpha,T}(\mathbf{u}_r[n])$, $\alpha \in \{1, 2\}$, can be written as (42), shown at the bottom of the next page, where $f_{\alpha}(\mathbf{u}_r^{(i)}[n])$ can be written as

$$f_{\alpha}(\mathbf{u}_r^{(i)}[n]) = \log_2 \left(g_{\alpha}(\mathbf{u}_r^{(i)}[n]) \right), \quad \forall k, n, \quad (43)$$

where $g_1(\mathbf{u}_r^{(i)}[n])$ and $g_2(\mathbf{u}_r^{(i)}[n])$ are written as (44a) and (44b), shown at the bottom of the next page, and $\nabla g_1(\mathbf{u}_r^{(i)}[n])$ and $\nabla g_2(\mathbf{u}_r^{(i)}[n])$ can be written as (45a) and (45b), shown at the bottom of the next page, where

$$a_{u_1, u_2}[n] = \left(\sqrt{\frac{\tilde{G}[n]}{\tilde{G}[n] + 1}} + \sqrt{\frac{1}{\tilde{G}[n] + 1}} \tilde{g}_{u_1, u_2}[n] \right)^4, \quad (46)$$

where $(u_1, u_2) \in \{(k, r), (s, r), (j, r)\}$.

Then, the problem (P4) can be reformulated as

$$\begin{aligned} \text{(P4.1)} : \quad & \max_{\{\mathbf{u}_r[n], \theta_{1,k}[n], \theta_{2,k}[n]\}} \theta \\ & \text{s.t. (1), (25b), (25d)-(25f), (40), (41).} \end{aligned} \quad (47)$$

Note that (P4.1) is typically convex, which can be solved by CVX.

The original problem (P1) can be solved by iteratively solving the problems (P3) and (P4) to approach the global optimal solution, in which the variables are optimized by updating in an alternating manner as shown in Algorithm 1.

Algorithm 1 BCD-Based Joint Optimization Algorithm

- 1: **Init:** $\mathcal{V}^{(0)}[n] = \{\xi_k^{(0)}[n], p_k^{(0)}[n], p_r^{(0)}[n], l_{k,\text{loc}}^{(0)}[n], \mathbf{u}_r^{(0)}[n]\}$, set $\varepsilon > 0$.
- 2: **repeat**
- 3: By $\mathcal{V}^{(i)}[n] \setminus \xi_k^{(i)}[n]$, solve the problem (P3.1), obtain and update time allocation $\xi_k[n]$.
- 4: By $\mathcal{V}^{(i)}[n] \setminus p_k^{(i)}[n]$, solve the problem (P3.2), obtain and update MD_k transmit power $p_k[n]$.
- 5: By $\mathcal{V}^{(i)}[n] \setminus p_r^{(i)}[n]$, solve the problem (P3.3), obtain and update UAV_r transmit power $p_r[n]$.
- 6: By $\mathcal{V}^{(i)}[n] \setminus l_{k,\text{loc}}^{(i)}[n]$, solve the problem (P3.4), obtain and update MD_k local computation allocation $l_{k,\text{loc}}[n]$.
- 7: By $\mathcal{V}^{(i)}[n] \setminus \mathbf{u}_r^{(i)}[n]$, solve the problem (P4), obtain and update UAV_r trajectory $\mathbf{u}_r[n]$.
- 8: Update $i \leftarrow i + 1$.
- 9: **until** The accuracy ε is achieved or i reaches I_1 .
- 10: **Output:** $\xi_k[n], p_k[n], p_r[n], l_{k,\text{loc}}[n], \mathbf{u}_r[n]$.

the problem as

$$(P5) : \max_{\{\xi_k[n], p_k[n], p_r[n], l_{k,\text{loc}}[n], \mathbf{u}_r[n]\}} Q_k^u \quad (48a)$$

$$\text{s.t. (1), (2), (5), (6), (17), (18), (22),}$$

$$l_{k,\text{loc}}[n] + \frac{1}{2} W \xi_k[n] \varphi_t \Phi_{k,\text{sec}} \geq Q_k^u, \quad \forall k, n. \quad (48b)$$

Through solving the problem (P5), we can obtain Q_k^u . Subsequently, we can assess the feasibility of Algorithm 1 and adjust the parameter initialization accordingly to ensure better performance [29].

In each iteration of the proposed algorithm, there are two standard convex optimization solutions and three SCA techniques, and the number of variables involved is $4KN + N$. Taking I_1 as the iteration number of BCD algorithm, we have the complexity of Algorithm 1 given as $I_1 O((KN)^{3.5} \log(1/\varepsilon))$ [37].

V. NUMERICAL RESULTS

In this section, we first evaluate the convergence of the proposed algorithm. Then, the impact of different constraints on the max-min secure computing capacity is compared. Lastly, we conduct a comparison between the proposed algorithm and four benchmark approaches. Some parameters are provided in Table II, and others are as follows [5], [31].

C. Feasibility and Complexity of Algorithm 1

The required Q_k may not satisfy the initialization parameters on the initial iteration. To make the problem solvable, we first check the feasibility of Algorithm 1 by optimizing

$$f_{\alpha, T}(\mathbf{u}_r[n]) = f_{\alpha}(\mathbf{u}_r^{(i)}[n]) + \nabla f_{\alpha}(\mathbf{u}_r^{(i)}[n]) (\mathbf{u}_r[n] - \mathbf{u}_r^{(i)}[n]) = f_{\alpha}(\mathbf{u}_r^{(i)}[n]) + \frac{\nabla g_{\alpha}(\mathbf{u}_r^{(i)}[n])}{g_{\alpha}(\mathbf{u}_r^{(i)}[n]) \ln 2} (\mathbf{u}_r[n] - \mathbf{u}_r^{(i)}[n]), \quad \forall k, n \quad (42)$$

$$g_1(\mathbf{u}_r^{(i)}[n]) = n_r^2 n_s^2 d_{j,r}^2[n] d_{k,r}^2[n] d_{r,s}^2[n] + a_{j,r}[n] n_s^2 P_j \beta_0 d_{k,r}^2[n] d_{r,s}^2[n] + a_{k,r}[n] n_s^2 p_k[n] \beta_0 d_{j,r}^2[n] d_{r,s}^2[n] + a_{r,s}[n] n_r^2 p_r[n] \beta_0 d_{j,r}^2[n] d_{k,r}^2[n] + a_{k,r}[n] a_{r,s}[n] p_k[n] p_r[n] \beta_0^2 d_{j,r}^2[n], \quad \forall k, n, \quad (44a)$$

$$g_2(\mathbf{u}_r^{(i)}[n]) = n_r^2 n_s^2 d_{j,r}^2[n] d_{k,r}^2[n] d_{r,s}^2[n] + a_{j,r}[n] n_s^2 P_j \beta_0 d_{k,r}^2[n] d_{r,s}^2[n] + a_{k,r}[n] n_s^2 p_k[n] \beta_0 d_{j,r}^2[n] d_{r,s}^2[n] + a_{r,s}[n] n_r^2 p_r[n] \beta_0 d_{j,r}^2[n] d_{k,r}^2[n], \quad \forall k, n \quad (44b)$$

$$\begin{aligned} \nabla g_1(\mathbf{u}_r^{(i)}[n]) &= 2 n_r^2 n_s^2 \left((\mathbf{u}_r^{(i)}[n] - \mathbf{v}_j) d_{k,r}^2[n] d_{r,s}^2[n] + d_{j,r}^2[n] (\mathbf{u}_r^{(i)}[n] - \mathbf{v}_k) d_{r,s}^2[n] + d_{j,r}^2[n] d_{k,r}^2[n] (\mathbf{u}_r^{(i)}[n] - \mathbf{v}_s) \right) \\ &\quad + 2 a_{j,r}[n] n_s^2 P_j \beta_0 \left((\mathbf{u}_r^{(i)}[n] - \mathbf{v}_k) d_{r,s}^2[n] + d_{k,r}^2[n] (\mathbf{u}_r^{(i)}[n] - \mathbf{v}_s) \right) \\ &\quad + 2 a_{k,r}[n] n_s^2 p_k[n] \beta_0 \left((\mathbf{u}_r^{(i)}[n] - \mathbf{v}_j) d_{r,s}^2[n] + d_{j,r}^2[n] (\mathbf{u}_r^{(i)}[n] - \mathbf{v}_s) \right) \\ &\quad + 2 a_{r,s}[n] n_r^2 p_r[n] \beta_0 \left((\mathbf{u}_r^{(i)}[n] - \mathbf{v}_j) d_{k,r}^2[n] + d_{j,r}^2[n] (\mathbf{u}_r^{(i)}[n] - \mathbf{v}_k) \right) \\ &\quad + 2 a_{k,r}[n] a_{r,s}[n] p_k[n] p_r[n] \beta_0^2 (\mathbf{u}_r^{(i)}[n] - \mathbf{v}_j), \quad \forall k, n, \end{aligned} \quad (45a)$$

$$\begin{aligned} \nabla g_2(\mathbf{u}_r^{(i)}[n]) &= 2 n_r^2 n_s^2 \left((\mathbf{u}_r^{(i)}[n] - \mathbf{v}_j) d_{k,r}^2[n] d_{r,s}^2[n] + d_{j,r}^2[n] (\mathbf{u}_r^{(i)}[n] - \mathbf{v}_k) d_{r,s}^2[n] + d_{j,r}^2[n] d_{k,r}^2[n] (\mathbf{u}_r^{(i)}[n] - \mathbf{v}_s) \right) \\ &\quad + 2 a_{j,r}[n] n_s^2 P_j \beta_0 \left((\mathbf{u}_r^{(i)}[n] - \mathbf{v}_k) d_{r,s}^2[n] + d_{k,r}^2[n] (\mathbf{u}_r^{(i)}[n] - \mathbf{v}_s) \right) \\ &\quad + 2 a_{k,r}[n] n_s^2 p_k[n] \beta_0 \left((\mathbf{u}_r^{(i)}[n] - \mathbf{v}_j) d_{r,s}^2[n] + d_{j,r}^2[n] (\mathbf{u}_r^{(i)}[n] - \mathbf{v}_s) \right) \\ &\quad + 2 a_{r,s}[n] n_r^2 p_r[n] \beta_0 \left((\mathbf{u}_r^{(i)}[n] - \mathbf{v}_j) d_{k,r}^2[n] + d_{j,r}^2[n] (\mathbf{u}_r^{(i)}[n] - \mathbf{v}_k) \right), \quad \forall k, n \end{aligned} \quad (45b)$$

TABLE II
PARAMETER SETTING

Parameters	Values
Channel power gain at a unit distance	$\beta_0 = -60$ dB
Altitude of UAVs	$H_r = H_e = 100$ m
The power ratio of signal matrix	$\tilde{G}[n] = 30$
Length of the time slot	$\varphi_t = 0.5$ s
Received noise power	$n_r = n_e = n_s = -110$ dBm
CPU cycles required per bit computing	$c_k = c_s = 10^3$ cycles/bit
Maximum CPU frequency	$F_k^{\max} = 10^9$ Hz, $F_s^{\max} = 10^{12}$ Hz
Communication bandwidth	$W = 1$ MHz
MD _k CPU capacity coefficient	$k_k = 10^{-27}$
MD _k power budget	$P_k^{\text{ave}} = 1$ W

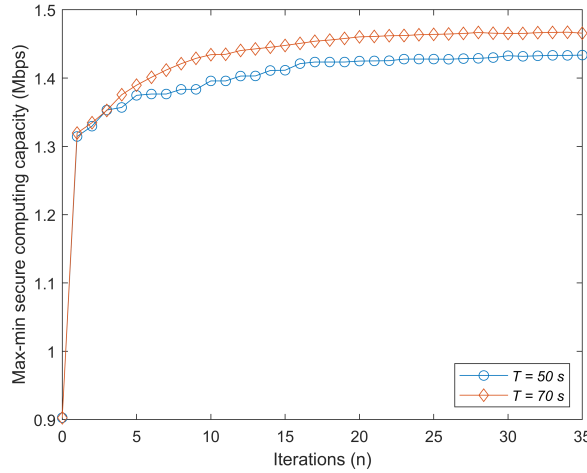


Fig. 2. Convergence of Algorithm 1 with different values of T , where $P_k^{\max} = 0.2$ W, $P_r^{\max} = 0.5$ W and $P_j = 0.5$ W.

Within a sea area of 1500×2000 m², we deploy five MDs in our proposed scheme. The locations of CES and CJ are both fixed at $[0, 0]^T$. UAV_r flies from $\mathbf{u}_r^I = [0, 0]^T$ to $\mathbf{u}_r^F = [2000, 0]^T$, and UAV_e flies from $\mathbf{u}_e^I = [0, 750]^T$ to $\mathbf{u}_e^F = [2000, -750]^T$, in meters. The maximum speed of UAV_r is set as $V_r^{\max} = 50$ m/s. The minimum collision avoidance distance between UAVs is set as $d_{\min} = 1$ m. UAV_e flies along a straight line with a constant speed.

Fig. 2 shows the convergence of our proposed algorithm while considering different values of T . In Fig. 2, it becomes apparent that an increase in the value of T results in a corresponding increase in the max-min secure computing capacity. This occurrence can be attributed to the fact that as T grows larger, the UAV_r can be afforded more time to approach the MDs, enabling it to deliver improved service quality. The extended duration allows the UAV_r to establish closer proximity with the MDs. Thus, enhancing the overall performance and contributing to the higher secure computing capacity.

Fig. 3 and Fig. 4 show the optimized trajectories of UAV_r in our proposed scheme with different T . Meanwhile, the given

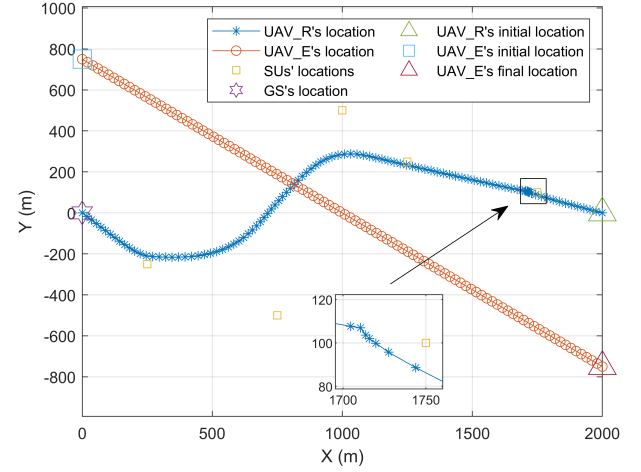


Fig. 3. Optimized trajectories for UAV_r vs. T , where $P_k^{\max} = 0.1$ W, $P_r^{\max} = 0.5$ W, $P_j = 0.5$ W, and $T = 50$ s.

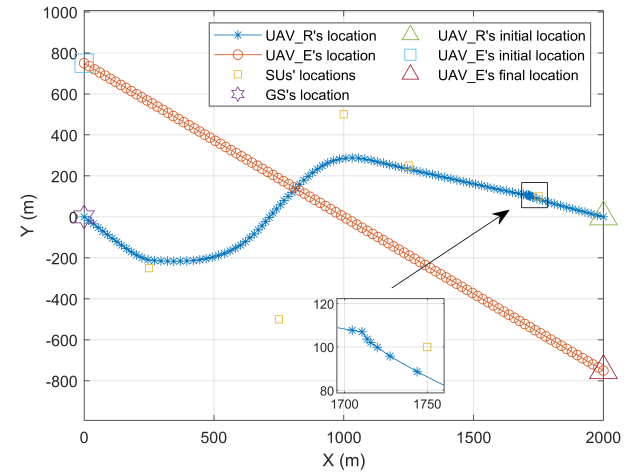


Fig. 4. Optimized trajectories for UAV_r vs. T , where $P_k^{\max} = 0.1$ W, $P_r^{\max} = 0.5$ W, $P_j = 0.5$ W, and $T = 70$ s.

trajectory of UAV_e is also shown. In the case of $T = 50$ s as shown in Fig. 3, because of the short time, UAV_r almost passes over a few MDs during the flight. With longer time $T = 70$ s as shown in Fig. 4, UAV_r can pass over more MDs, obtaining more time slots staying above each MD. Thus, the hovering time over each MD will be longer, resulting in better performance as illustrated in Fig. 6 and Fig. 7.

Fig. 5 illustrates the optimized speeds for UAV_r with different T . As shown in Fig. 3 and Fig. 4, UAV_r will pass over as many MDs as possible, and hover over each MD as long as possible. It results that UAV_r flies away from a MD as fast as possible, and slows down slowly when it approaches next MD. As shown in Fig. 5(a), there is a significant change in the speed of UAV_r between time slot 80 and 90, indicating that UAV_r stays above the corresponding MD for a longer period of time. And in Fig. 5(b), a similar change in speed occurs five times. It can be seen that UAV_r has more time to approach MDs to provide better service due to the increase T .

Fig. 6 illustrates the max-min secure computing capacity with different values of $P_k^{\max}$. It can be observed that with the increase of $P_k^{\max}$, the max-min secure computing capacity is

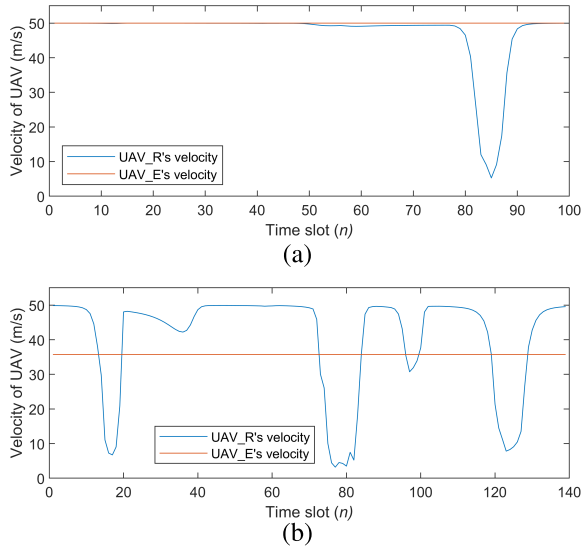


Fig. 5. Optimized flight speeds for UAV_r with different T , where $P_k^{\max} = 0.1$ W, $P_r^{\max} = 0.5$ W and $P_j = 0.5$ W. (a) $T = 50$ s, (b) $T = 70$ s.

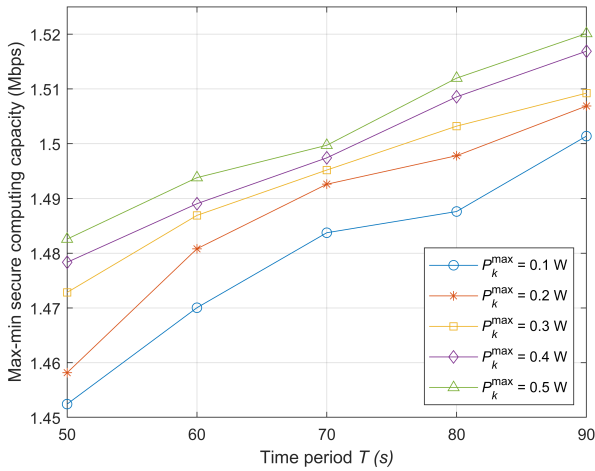


Fig. 6. Secure computing capacity with different $P_k^{\max}$, where $P_r^{\max} = 0.8$ W and $P_j = 0.5$ W.

becoming larger. This is because with higher transmit power, MDs will have more energy to offload computing tasks to CES through UAV_r, which has more powerful computing ability.

Fig. 7 shows the max-min secure computing capacity with different values of $P_r^{\max}$. When $P_r^{\max}$ becomes larger, UAV_r can forward more information to CES with larger information rate. While CJ only transmits the jamming signal in the first phase of each time slot. Thus, for the information forwarded by UAV_r to CES, the effective transmission rate increases with larger $P_r^{\max}$, resulting in a better performance.

Fig. 8 shows the max-min secure computing capacity with different values of P_j . It can be observed that with increase of P_j , the max-min secure computing capacity is becoming larger. This is because CES knows the interference signal emitted by CJ in advance, while UAV_e does not. When P_j increases, producing stronger interference to the UAV_e, resulting in a smaller eavesdropping rate.

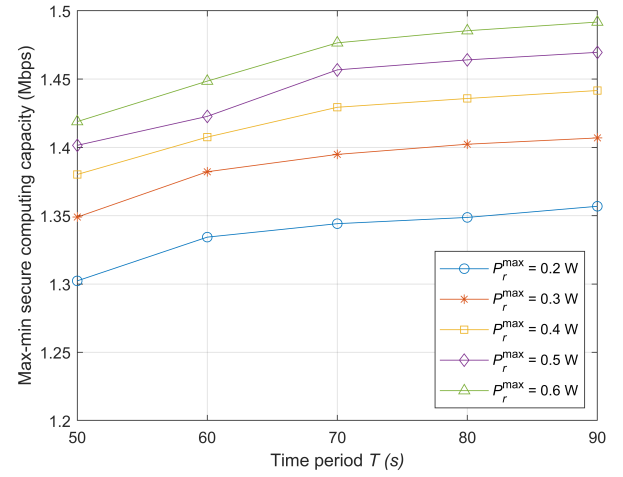


Fig. 7. Secure computing capacity with different $P_r^{\max}$, where $P_k^{\max} = 0.1$ W and $P_j = 0.5$ W.

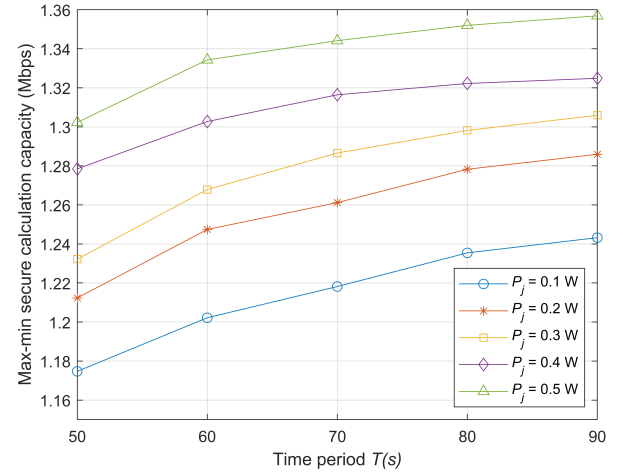


Fig. 8. Secure computing capacity with different P_j , where $P_k^{\max} = 0.1$ W and $P_r^{\max} = 0.2$ W.

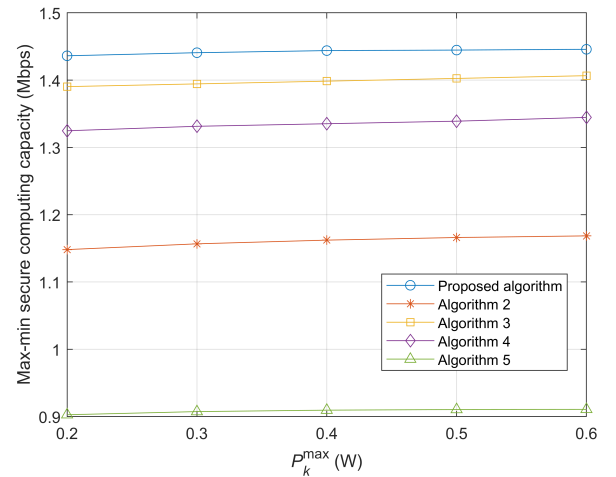


Fig. 9. Secure computing capacity with different $P_k^{\max}$, where $T = 50$ s, $P_r^{\max} = 0.5$ W and $P_j = 0.5$ W. Algorithm 2 - 5 are four benchmarks.

Fig. 9 and Fig. 10 illustrate the secure computing capacity comparisons of our proposed algorithm with four benchmarks with the same target of maximizing secure computing capacity

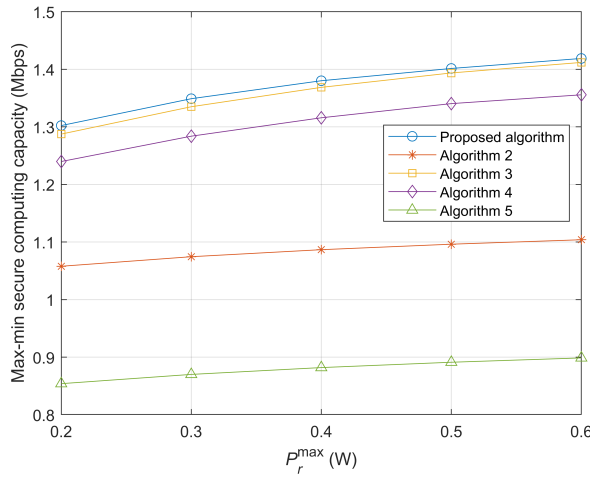


Fig. 10. Secure computing capacity with different $P_r^{\max}$, where $T = 50$ s, $P_k^{\max} = 0.1$ W and $P_j = 0.5$ W. Algorithm 2 - 5 are four benchmarks.

of the system. The specific descriptions of four benchmarks are as follows.

Algorithm 2: The time slot allocation factor $\xi_k[n]$ is fixed, while MD_k transmit power $p_k[n]$, UAV_r transmit power $p_r[n]$, MD_k local computation allocation $l_{k,loc}[n]$, and UAV_r trajectory $\mathbf{u}_r[n]$ are optimized.

Algorithm 3: MD_k transmit power $p_k[n]$ is fixed, while the time slot allocation factor $\xi_k[n]$, MD_k local computation allocation $l_{k,loc}[n]$, UAV_r transmit power $p_r[n]$, and UAV_r trajectory $\mathbf{u}_r[n]$ are optimized.

Algorithm 4: UAV_r transmit power $p_r[n]$ is fixed, while the time slot allocation factor $\xi_k[n]$, MD_k local computation allocation $l_{k,loc}[n]$, MD_k transmit power $p_k[n]$, and UAV_r trajectory $\mathbf{u}_r[n]$ are optimized.

Algorithm 5: UAV_r trajectory $\mathbf{u}_r[n]$ is fixed, while the time slot allocation factor $\xi_k[n]$, MD_k local computation allocation $l_{k,loc}[n]$, MD_k transmit power $p_k[n]$, and UAV_r transmit power $p_r[n]$ are optimized.

By comparing the secure computing capacity of the proposed algorithm with four benchmarks having different $P_k^{\max}$ values in Fig. 9, it is evident that the proposed algorithm is better. The proposed algorithm outperforms four benchmarks with different values of $P_k^{\max}$. Furthermore, the proposed algorithm optimizes the time slot allocation factor when it is compared to Algorithm 2, highlighting the importance of this factor in improving performance. In addition to optimizing the transmit power of MD_k , the proposed algorithm significantly improves performance compared to Algorithm 3. Furthermore, the proposed algorithm optimizes the transmit power of UAV_r to enhance performance, a factor not considered in Algorithm 4. Finally, compared to Algorithm 5, the proposed algorithm also optimizes the UAV_r trajectory, indicating the significance of this factor in improving performance.

In Fig. 10, it is evident that the proposed algorithm outperforms four benchmarks with different values of $P_r^{\max}$. It can be observed that Algorithm 2 achieves a lower performance in secure offloading rate as it does not optimize the time slot allocation factor. Similarly, Algorithm 3 achieves a lower performance in secure offloading rate as it does not

optimize MD_k transmit power. Algorithm 4 achieves a lower performance in secure offloading rate as it does not optimize UAV_r transmit power. Finally, Algorithm 5 achieves a lower performance in secure offloading rate as it does not optimize UAV_r trajectory. This highlights the significance of optimizing time slot allocation factor, MD_k and UAV_r transmit powers, and UAV_r trajectory in enhancing the overall performance of the proposed algorithm.

VI. CONCLUSION

In this paper, a secure communication scheme was proposed to enhance the secure computing capacity performance for the UAV-relay-assisted maritime MEC, where UAV_r helps to forward the offloading tasks of MDs to CES. To reduce UAV_e eavesdropping, CJ generates the interfering signals. The max-min secure computing capacity of MDs is achieved by optimizing the UAV trajectory, time slot allocation, transmit power and computation allocation with the constraints of UAV_r flight speed, UAV anti-collision, UAV_r transmit power, MDs transmit power, MDs local computation ability, MDs computing task requirements and CES CPU frequency. The formulated OP is non-convexity because of the coupling of multiple variables. We first transform OP by introducing auxiliary variables. Then, the transformed OP is decomposed and solved in an iterative manner by applying BCD and SCA. The simulation results show that the proposed algorithm can achieve better secure computing capacity performance compared with four benchmarks.

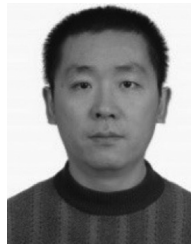
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